

TCP Solved Problems

Rules:

SL	Rule
R1	SYN and SYN+ACK segments have 1 byte each.
R2	Except for the SYN segment, virtually all TCP segments have a valid ACK.
R3	A pure ACK segment (ACK without data) does not consume a sequence number.
R4	The sequence number (SEQ.) is the byte number of the first byte of a number.
R5	The FIN segment consumes a sequence number. This means it is assumed to have 1 byte of data.
R6	Although SYN and SYN-ACK segments are assumed to have one byte of data each, these bytes are not stored by the receiver, as they are not user data.
R7	The SEQ of a segment from device A depends on its own ISN and the number of bytes it has sent. Conversely, the ACK of that segment depends on the ISN of the other party (B) and the number of the last byte A has received in a continuous stream without any gaps.
R8	If the SEQ of C1 is P and C1 carries D bytes of data, the last byte's number of the segment is $SEQ + D - 1 = P + D - 1$
R9	If the SEQ of C1 is P and C1 carries D bytes of data, the SEQ of the C2 bytes is $P + D$ [if everything goes well]
R10	If the SEQ of C1 is P and C1 carries D bytes of data, the acknowledgement number (ACK) of the next segment from the server is $ACK = SEQ + D = P + D$
R11	For selected Repeat & Go-Back-N, if a device's rwnd is R and it has received three segments with D1, D2, and D3 bytes successfully without any missing bytes, the resulting rwnd will be $R - D1 - D2 - D3$.
R12	For selected Repeat method, if a device's rwnd is R and it has received three segments with D1, and D3 bytes successfully while D2 bytes between D1 and D3 are missing, the resulting rwnd will be $R - D1 - D3$.
R13	For the Go-Back_N method, if a device's rwnd is R and it has received three segments with D1, and D3 bytes successfully, while D2 bytes between D1 and D3 are missing, the resulting rwnd will be $R - D1$.

Problem#1

Suppose that the ISNs of the client and servers are 500 and 265, respectively. C1, C2, and C3 represent the client segments, while S1 and S2 represent the server segments (see Fig. 1). Answer the following questions:

- What are the sequence and acknowledgement number of the C1 segment?
- How much data is sent in the S1?
- What are the sequence and acknowledgement numbers of the S1 segment?
- What is the last byte number of the C2 segment?
- What are the sequence and acknowledgement numbers of the C3 segment?
- What are the sequence and acknowledgement numbers of the S2 segment?

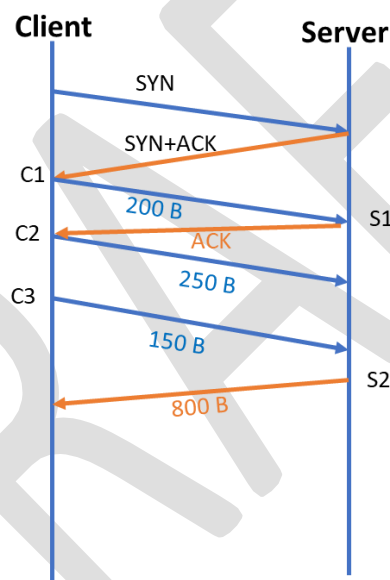


Fig. 1

Solution:

Let N_k represents the number of bytes in the segment k.

- SEQ No.

Client's ISN is $ISN_c = 500$, and SYN segment carries 1 (call it $N_{SYN} = 1$) byte of data.

Hence, SEQ of C1: $ISN_c + N_{SYN} = 500 + 1 = 501$.

Since the ISN of the server is 265 and SYN+ACK carries 1 byte of data, the ACK of the C1 segment is $265 + 1 = 266$.

- Since S1 carries only ACK, it has no data

c) SEQ No.

SEQ of S1 is $ISN_S + N_{SYN-ACK} = 265 + 1 = 266$, as this is the first segment after the SYN+ACK segment, which had 1 byte of data.

ACK No.

Up to now, the server has received two segments, SYN and C1, which carry 1 and 200 bytes of data, respectively, without any missing bytes. ACK number of S1 is $ISN_C + N_{SYN} + N_{C1} = 500 + 1 + 200 = 701$.

d) Including C2, the client has sent three segments (SYN, C1, and C2) with 1, 200, and 250 bytes of data. Hence, the last byte's number of the C2 segment is

$$ISN_C + N_{SYN} + N_{C1} + N_{C2} - 1 = 500 + 1 + 200 + 250 - 1 = 950 \text{ [R7]}$$

e) Sequence Number:

So far, the client has sent three segments: SYN, C1, and C2 with 1, 200, and 250 bytes, respectively. Hence, its SEQ is

$$ISN_C + N_{SYN} + N_{C1} + N_{C2} = 500 + 1 + 200 + 250 = 951$$

ACK number:

Before sending C3, the client had received two segments: SYN+ACK and S1 with 1 and 0 bytes of data, respectively, without any missing bytes.

Hence, the ACK number of C3 is

$$ISN_S + N_{SYN-ACK} + N_{S1} = 265 + 1 + 0 = 266$$

d) Sequence No.

$$\text{SEQ of S2} = ISN_S + N_{SYN-ACK} + N_{S1} = 265 + 1 + 0 = 266$$

ACK No.

So far, the server has received four segments without any missing bytes.

$$\begin{aligned} \text{ACK of S2} &= ISN_C + N_{SYN} + N_{C1} + N_{C2} + N_{C2} = 500 + 1 + 200 + 250 + 150 = 1101 \\ &= 500 + 1 + 200 + 250 + 150 = 1101 \end{aligned}$$

Problem#2

Suppose that the SEQ, ACK, and receive window size (rwnd) of the C1 segment are 1204, 3210, and 930, respectively (see Fig. 2). Assume that the rwnd size advertised in S1 is 1500.

- Find the SEQ, ACK, and rwnd size of the C3 segment.
- Find the SEQ, ACK, and rwnd size of the S2 segment.
- Find the ACK of the S3 segment.
- Is it possible that the rwnd size advertised in S3 is 820?
- After the FIN segment, C6, there is still a data transmission of 50 B. Is it valid?

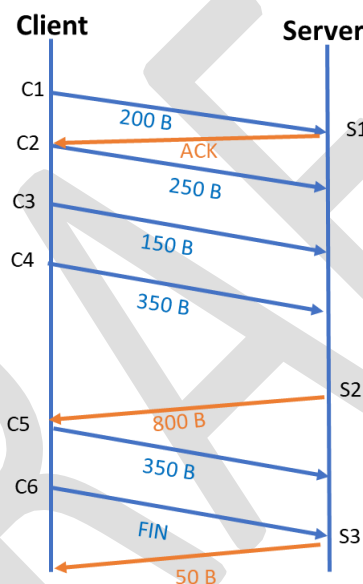


Fig. 2

Solution:

ISNs of the client and servers are not given. Though C1's SEQ and ACK are given, the SEQ of the server is not given. We can calculate this as follows:

C1's ACK is 3210. This means, the client has received all bytes up to the byte number 3209 and now it is requesting data from the byte number 3210. Thus, next time, the server has to send from the byte number 3210. So, the SEQ of S1 is 3210.

Hence, SEQ of C1 is 1204 while that of S1 is 3210.

- For C3:

Sequence number:

Before C3, the client sent two segments that carried 200 and 250 bytes, respectively.

SEQ of C2: $SEQ\ of\ C1 + N_{C1} + N_{C2} = 1204 + 200 + 250 = \mathbf{1654}$

ACK:

The previous segment of C2 is C1, which had the ACK number of 3210. After C1, the client has received only one segment, S1, from the server before sending C3. However, this ACK segment carries no data. Hence, its ACK remains the same: 3210.

Another method: the ACK of the C2 is SEQ of S1+0 = 3210+0=**3210**.

rwnd:

At the beginning (when C1 is sent in this example), the rcwnd size was 930. This means, the client has memory to store 930 bytes. Before sending C2, it has received only one segment, S1, from the server and S1 carries no data. Hence, no new data has been stored on its receive window due to the reception of S1. Thus, the client's rwnd size remains the same. Thus, rwnd is **930**.

b) SEQ:

SEQ of S1 was 3210 and it had no data. Thus, SEQ of S2 is SEQ of S1+ N_{S1} = 3210 + 0 = 3210

ACK:

Before sending S2, the server received four segments C1, C2, C3, and C4 that carried 200 B, 250 B, 150 B, and 320 B, respectively.

ACK:

$SEQ\ of\ C1 + N_{C1} + N_{C2} + N_{C3} + N_{C4} = 1204 + 200 + 250 + 150 + 350 = 2154$

rwnd:

The rwnd size was 1500 during S1 transmission. Hence, it has memory to store 1500 bytes more data. Between S1 and before S2, the server has received three segments C2, C3, and C4 that carried 250 B, 150 B, and 320 B, respectively. No information is given whether the process consumed any bytes during this time. Hence, we assume that the process did not consume any bytes. The remaining space in the server's receive window is

$$rwnd\ in\ S1 - N_{C2} - N_{C3} - N_{C4} = 1500 - 250 - 150 - 350 = 750$$

c) Only FIN segment carries 1 byte of data.

ACK of S3:

$$SEQ\ of\ C1 + N_{C1} + N_{C2} + N_{C3} + N_{C4} + N_{C5} + N_{C6}$$

$$= 1204 + 200 + 250 + 150 + 350 + 350 + 1 = 2505$$

Another method:

In question b, we found that the ACK of the S2 was 2154.

After sending S2, the server has received C5 and C6. Hence,

$$ACK\ of\ S3 = ACK\ of\ S2 + N_{C5} + N_{C6} = 2154 + 350 + 1 = 2505$$

d) Yes, possible. Let's justify.

In S2, the rwnd size of the server was 550. Thereafter, it has received 350 bytes more. Though the FIN segment is assumed to carry 1 byte, it is not stored as this is not user data.

Then, in S3, the rwnd size should be $550 - 350 = 200$.

However, we are told that S3 advertises rwnd of 820. This means after sending S2, the process has consumed $820 - 200 = 620$ bytes. Thus, this 620-byte space becomes free. Thus, total free space becomes $200 + 620 = 820$ bytes.

e) Yes, it is valid. Since the FIN segment was sent by the client, it can no longer send any data. However, the server can. This is a half-close connection.

Problem#3

Suppose that the ISNs of the client and servers are 500 and 265, respectively (see Fig. 3). The rwnd size of the client and server are 900 and 720, respectively.

- Find the ACK in C1 and C2 segments.
- Find the SEQ and data size in S5. How many bytes are sent in S5?
- If the advertised rwnd in C1 segment is 750, which sliding window protocol is used here?
- What are the ACK and rwnd size in C3?
- Find SEQ, ACK, rwnd, and data size in C4 segment.

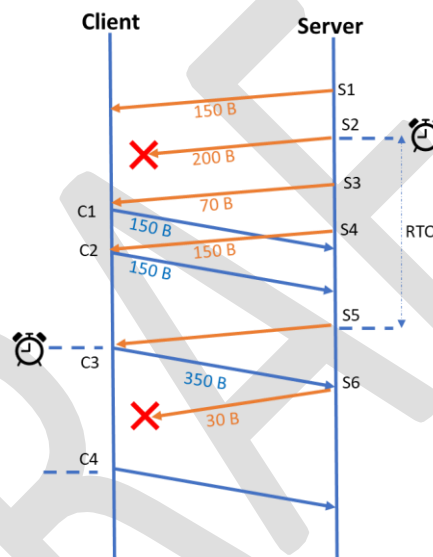


Fig. 3

- To find the ACK of the client's segment, we have to consider the sequence number of the server's segment. The server's ISN is 265, while S1 is the first data transmission from the server. Before the data transmission, there was a 3-way handshaking procedure for the TCP connection (which is not shown in Fig. 3). The SYN-ACK is assumed to have one byte of data. Before sending C1, the client received two segments: SYN-ACK and S1; S2 was lost, and S3 will not be acknowledged because the bytes in S2 are still missing.

Thus, the ACK number of the C1 segment is

$$ISNs + N_{SYN-ACK} + N_{S1} = 265 + 1 + 150 = 416$$

C2 will have the same ACK as the C1 segment even after getting S4 because the bytes in the S2 segments are still missing. Hence the ACK of C2 is 416.

- b) After sending every segment, the sender sets a timer. In Fig. 1, this case is shown for S2 and C3. S2 segment lost. After the timer expires, the S2 segment is resent as S5. Thus, the SEQ of S5 will be the same as that of S2. The SEQ of S2 is

$$ISN_S + N_{SYN-ACK} + N_{S1} = 265 + 1 + 150 = 416$$

Since S5 is the same as S2, it will contain 200 B as S2.

- c) Before sending C1, the client has received two segments, S1 and S3, successfully. Its initial $rwnd$ was 900. After receiving these two segments, the $rwnd$ in C1 should have been $rwnd_C - N_{S1} - N_{S3} = 900 - 150 - 70 = 680$.

However, the given $rwnd$ in C1 is 750, which can be obtained as

$$rwnd_C - N_{S1} = 900 - 150 = 750$$

This means the client did not store the data in S3 because the bytes in S2 have yet to be received. If a segment is missing and the subsequent received segments' bytes are not stored, the corresponding TCP sliding window method is called the Go-Back-N method.

- d) Since the Go-Back-N method is used, no segments after the missing segment is stored. Hence, the ACK number of the C3 segment is

$$ISN_S + N_{SYN-ACK} + N_{S1} + N_{S5} = 265 + 1 + 150 + 200 = 616$$

The $rwnd$ size advertised in C3 is

$$rwnd_C - N_{S1} - N_{S5} = 900 - 150 - 200 = 550$$

- e) SEQ

C1 and C2 segments are acknowledged in S5 while C3's ACK is lost as S6 is lost. Thus, after RTO for C3, C3 will be retransmitted. Thus, C4 is actually C3. To compute SEQ of C4, we have to calculate the SEQ of C3, which is given below:

$$ISN_C + N_{SYN} + N_{C1} + N_{C2} = 500 + 1 + 150 + 150 = 801$$

ACK

Since C4 contains the same info as the C3, it's ACK is 616 (see Q.d)

$rwnd$ size

Since C4 contains the same info as the C3, it's $rwnd$ is 550 (see Q.d)

Problem#4

In a **selective repeat** TCP connection, the client and server have the following values and flow:

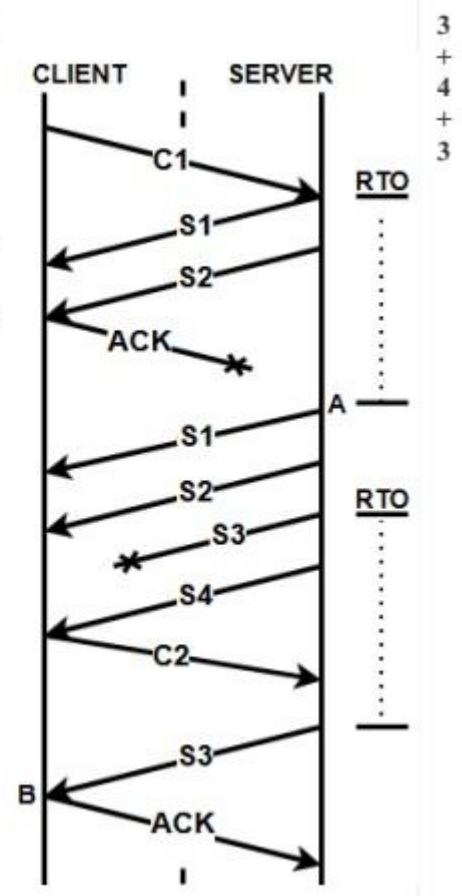
ISN of Client = **6729**, RWND of Client = **3200**;

ISN of Server = **6214**, RWND of Server = **8000**

The data size of the segments is given as shown in bytes:

C1=1421, C2=1320, S1=?, S2=443, S3=122 and S4=220

- I. If the **sequence** number of the **second S2** segment is **7000**, calculate the data size of S1.
- II. Calculate the **sequence** and the **acknowledgement** number of segment C2.
- III. Calculate the **rwnd** of the client at **point B**.



I.

$$ISN_S + N_{SYN-ACK} + N_{S1} = 7000$$

$$6214 + 1 + N_{S1} = 7000$$

$$N_{S1} = 7000 - 6214 - 1 = 785$$

II. SEQ:

Before C2, the client has sent only one segment: C1 with 1421 bytes.

SEQ of C2: $ISN_C + N_{SYN} + N_{C1} = 6729 + 1 + 1421 = 8151$

ACK:

Before sending C2, S1, S2, and S4 have been successfully received. S4 is stored even it arrives out of order because the Selective Repeat Sliding window method is being used. However, due to the missing bytes of S3, S4 will not be acknowledged. Thus, S2 will be acknowledged in C2.

ACK of C2:

$$ISN_S + N_{SYN-ACK} + N_{S1} + N_{S2} = 6214 + 1 + 785 + 443 = 7643$$

- III. At the beginning of the process, the client has 3200 bytes space in its buffer memory. Up to the point B, it has received S1-S4 four segments have been successfully received and stored. Thus, the remaining space in its buffer:

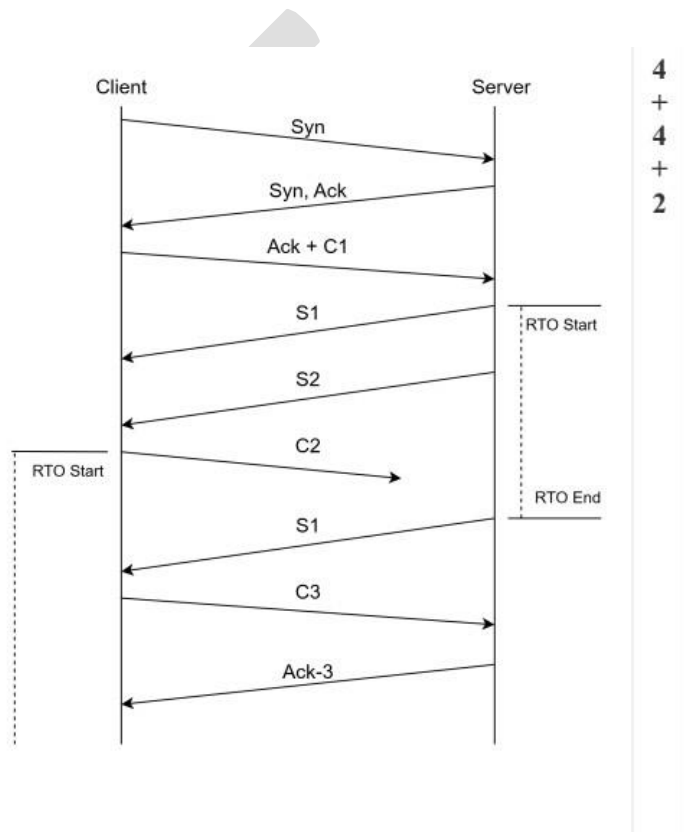
$$rwnd_c - N_{S1} - N_{S2} - N_{S3} - N_{S4} = 3200 - 785 - 433 - 122 - 220 = 1640$$

Problem#5

While transferring data, the ISN of the client is **1455**, and the ISN of the server is **2010**. Initially, the *rwnd* of the client is **8000** bytes, the *rwnd* of the server is **10000** bytes, and they are using **go-back-n** protocol to send data. The data size of the segments are given as shown in bytes:

C1= 320, C2=111, C3=260, S1=220, and S2=421.

- I. Explain why the server is resending segment S1 and what the client will do with segment S1.
- II. Calculate the **sequence** and **acknowledgment numbers** of the **C3** segment.
- III. Calculate the *rwnd* of the server after receiving the **C3** segment from the client.



Solution:

- I. After the 3-way handshaking, the server has sent two segments: S1 and S2. The client sends C2 with 111 bytes of data and ACK for both S1 and S2 (we call it a cumulative ACK as multiple segments are acknowledged in a segment). However, C2 is lost.

Within the RTO of S1, it has not been acknowledged. Hence, the server will assume that S1 could not reach the client. Hence, it will resend it after the RTO.

After receiving the resent S1, the client will discard the segment because it has already received and stored S1 during its first transmission. It will send ACK for this S1 segment.

- II. SEQ

Before sending C3, the client has sent 3 segments: SYN, ACK-C1, and C2. Although C2 is lost, the client will not consider the ACK in S1 because RTO has not expired yet. Thus, it will assume that the bytes in C2 have been sent. Thus, SEQ:

$$ISN_C + N_{SYN} + N_{ACK-C1} + N_{C2} = 1455 + 1 + 320 + 111 = 1887$$

ACK

Before sending C3, the client has received three segments from the server: SYN-ACK, S1, and S2.

$$ISN_S + N_{SYN-ACK} + N_{S1} + N_{S2} = 2010 + 1 + 220 + 421 = 2652$$

- III. Up to the reception of C3, the server has received three segments SYN, ACK-C1, and C3. However, C3 has arrived out of order because C2 is lost. Since the Go-Back-N sliding window method is used here, any out of order segments will not be stored. Thus, C3 will be discarded. Furthermore, the one byte in SYN is not stored as it is not user data. Hence, the *rwnd* size of the server becomes

$$rwnd_S - N_{ACK-C1} = 10000 - 320 = 9680$$